Deployment Team Take-Home Assignment Part C

Ng Zheng Xun

Part C – Opinion

Six months after deploying the system you designed in Part A, users report that answers are sometimes outdated or irrelevant even though the source documents are correct and up to date. In about 200 words, what are the first three things you'd investigate, and how specifically would you go about each one? Note that you should name the specific methods and not just the category.

1) **Investigate the Embedding model** and if the embedding model is trained on the type of documents that we are investigating. Some examples of failed queries could be investigated one-by-one, by checking their vector similarities against semantic similarity. If vector similarities are low despite high semantic similarity, the embedding model needs to change.

2) **Investigate the Retrieval model and algorithm** to investigate if it is indeed functioning correctly. Some sample queries could be examined. If the correct documents were indeed retrieved and in queue, but not surfaced up to be the top-few in queue, the retrieval algorithm needs to change.

3) **Investigate the Chunking process** is correct, based on our internal document's formatting and/or terminology. Some sample documents could be chunked and the chunks inspected manually. If there is an unnatural cut-off in the chunk sentences or if accidental headings/page numbers are inserted inside a sentence, the chunking process needs to change.

(151 words)